

SAMANTHA AN

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

B.S. Computer Engineering

- Relevant Coursework: Data Structures (CSE 30), Linear Algebra, Multivariable Calculus, Physics I

Los Angeles City College

Los Angeles, CA

Object-Oriented Programming in C++

EXPERIENCE

API Developer/Data Analyst Intern

Aug 2026 – Sept 2026

Magnakom

Los Angeles, CA

- Accelerated SellerCloud-to-WholeCell order matching, as measured by cutting per-order lookup time from 2–3 minutes to seconds (~90% reduction) across 50+ batches against ~19,000 records, by building a Python app with hash-map lookup across integrated REST APIs.
- Automated product data extraction in certain websites, as measured by a ~92% reduction in per-product collection time (hours to minutes), by building a Chrome plugin that exports listings directly to Google Sheets.

Intern

Mar 2026 – June 2026

UCSC KASA (Korean American Student Association)

Santa Cruz, CA

- Delivered a 250+ person campus event, as measured by on-time execution across all scheduled sessions, by coordinating scheduling, logistics, and staffing alongside fellow interns.

PROJECTS

AlphaTest – A Backtesting Trade Application | *Python, FastAPI, Pandas, React, Vite*

Sept 2026

- Engineered a Python/FastAPI backtesting engine covering 7 rule-based trading strategies (moving-average crossover, MACD, RSI, Bollinger Bands, and more), as measured by every strategy sharing one indicator library and parameter schema, by building a registry that auto-generates each strategy's UI form via dataclass introspection.
- Hardened the price-data ingestion pipeline against malformed requests, as measured by 6+ validation rules catching invalid tickers, future dates, and interval/lookback mismatches before any network call, by designing a typed request-validation layer on top of the yfinance library.
- Eliminated lookahead bias across all backtested strategies, as measured by shifting every signal one bar before it becomes a tradable position, by centralizing that lag logic in one shared signal-processing module reused by all 7 strategies

Writing Buddy | *Python, FastAPI, FAISS, Gemini API (AI Writing Agent)*

July 2026 – August 2026

- Personalized AI-generated writing to match individual style, as measured by stylistic fidelity across user writing samples, by building a Chrome extension with a Python/FastAPI backend powered by a Retrieval-Augmented Generation (RAG) pipeline.
- Improved retrieval accuracy for style-conditioned LLM output, as measured by relevance of retrieved passages, by designing an end-to-end RAG pipeline — chunking, Gemini API embedding, and FAISS indexing.
- Eliminated cross-user data leakage and uncontrolled billing exposure, as measured by remediation of a critical vulnerability before launch, by auditing a production-bound FastAPI service and replacing a shared global API client with a per-request architecture.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++, HTML/CSS

Frameworks & Libraries: Node.js, React, Vite

Tools & Platforms: REST APIs, Git, FAISS

Concepts: API security auditing, OAuth authentication, real-time systems, RAG pipelines & vector search

CERTIFICATIONS & HONORS

Code Academy C++ Certification of Completion (2026)

Deans Honor List (Fall 2025-Spring 2026)

Girls Who Code, Certified Graduate (2023) — completed a 2-week intensive program on cybersecurity fundamentals and industry practices.